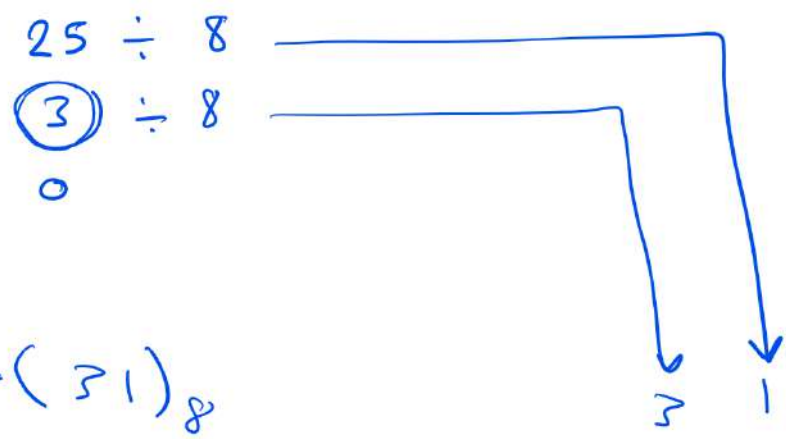


* Varigned Numbers

1. Decimal system : base = radix = 10 ✓
0, 1, . . . - 9
2. Binary system : base = radix = 2
0, 1,
3. Octal system : base = radix = 8
0, 1, . . . - 7
4. Hexadecimal " : base = radix = 16
0, 1, 2 . . . - 9, ¹⁰A, ¹¹B, ¹²C, ¹³D, ¹⁴E, ¹⁵F

Decimal → octal

$$(25)_{10} = (?)_8$$



Decimal $\rightarrow$ Hexadecimal

(2)

$$(25)_{10} = (?)_{16}$$

$$(25)_{10} = (19)_{16}$$

$$\begin{array}{r} 25 \div 16 \\ 1 \div 16 \\ 0 \end{array}$$

Arrows point from the remainders 1 and 9 to the right, indicating the hexadecimal digits 1 and 9.

$$(75)_{10} = (?)_{16}$$

$$(75)_{10} = (4B)_{16}$$

$$\begin{array}{r} 75 \div 16 \\ 4 \div 16 \\ 0 \end{array}$$

Arrows point from the remainders 4 and B to the right, indicating the hexadecimal digits 4 and B.

$$(25)_{10} = (?)_3 \quad \underline{\text{in base 3}}$$

$$25 \div 3$$

base 3 : 0, 1, 2

Octal $\rightarrow$ Decimal

$$(27)_8 = (?)_{10}$$

$$\begin{array}{c} 1 \quad 0 \\ 2 \quad 7 \end{array} = 2 \times 8^1 + 7 \times 8^0 = 16 + 7 = 23$$

$$(27)_8 = (23)_{10}$$

Hex $\rightarrow$ Decimal

⑤

$$(2F3A)_{16} = (?)_{10}$$

$$\begin{aligned} \overset{3}{2} \overset{2}{F} \overset{1}{3} \overset{0}{A} &= A \times 16^0 + 3 \times 16^1 + F \times 16^2 + 2 \times 16^3 \\ &= 10 + 3 \times 16 + 15 \times 256 + 2 \times 4096 \\ &= (12090)_{10} \end{aligned}$$

* Binary $\longleftrightarrow$ Octal

- each 3 bits in binary represents 1 digit in octal and vice versa.

$$(257)_8 = (?)_2$$

4 2 1 0 1 0	4 2 1 1 0 1	4 2 1 1 1 1
----------------	----------------	----------------

$$\begin{aligned} (257)_8 &= (010101111)_2 \quad \checkmark \\ &= (10101111)_2 \quad \checkmark \end{aligned}$$

$$(11101100)_2 = (?)_8$$

4 2 1 0 1 1 3	4 2 1 1 0 1 5	4 2 1 1 0 0 4
---------------------------	---------------------------	---------------------------

$$(11101100)_2 = (354)_8$$

(4)

Hex $\longleftrightarrow$ Binary

- each 4 bits in binary represents 1 digit in Hex and vice versa.

$$(C9B)_{16} = (?)_2$$

$\begin{array}{r} 8 \ 4 \ 2 \ 1 \\ 1 \ 1 \ 0 \ 0 \end{array}$

$\begin{array}{r} 8 \ 4 \ 2 \ 1 \\ 1 \ 0 \ 0 \ 1 \end{array}$

$\begin{array}{r} 8 \ 4 \ 2 \ 1 \\ 1 \ 0 \ 1 \ 1 \end{array}$

$$(C9B)_{16} = (110010011011)_2$$

$$(101110)_2 = (?)_{16}$$

$\begin{array}{r} 8 \ 4 \ 2 \ 1 \\ 0 \ 0 \ 1 \ 0 \\ \hline 2 \end{array}$

$\begin{array}{r} 8 \ 4 \ 2 \ 1 \\ 1 \ 1 \ 1 \ 0 \\ \hline E \end{array}$

$$(101110)_2 = (2E)_{16}$$

Hex $\longleftrightarrow$ Binary $\longleftrightarrow$ Octal

$$(3C)_{16} = (?)_8$$

$\begin{array}{r} 8 \ 4 \ 2 \ 1 \\ 0 \ 0 \ 1 \ 1 \\ \hline 3 \end{array}$

$\begin{array}{r} 8 \ 4 \ 2 \ 1 \\ 1 \ 1 \ 0 \ 0 \\ \hline 4 \end{array}$

$\longleftrightarrow$

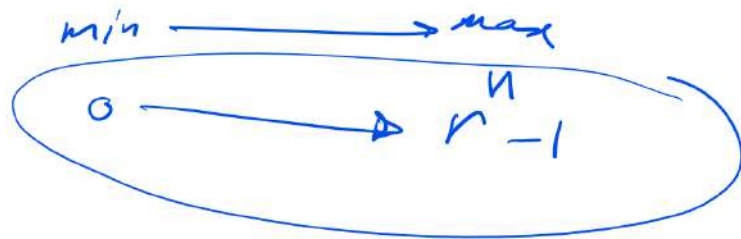
$\begin{array}{r} 8 \ 4 \ 2 \ 1 \\ 0 \ 0 \ 0 \\ \hline 0 \end{array}$
 $\begin{array}{r} 8 \ 4 \ 2 \ 1 \\ 1 \ 1 \ 1 \\ \hline 7 \end{array}$
 $\begin{array}{r} 8 \ 4 \ 2 \ 1 \\ 1 \ 0 \ 0 \\ \hline 4 \end{array}$

$$(3C)_{16} = (74)_8$$

* unsigned numbers:

(5)

Range of values:

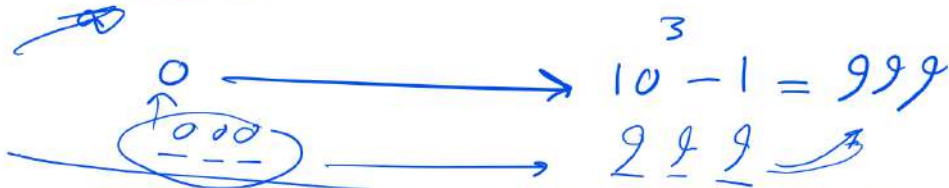


r = radix

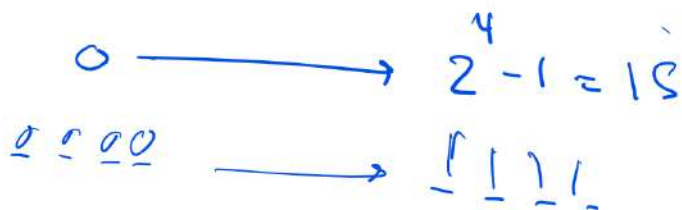
n = # of digits

Example:

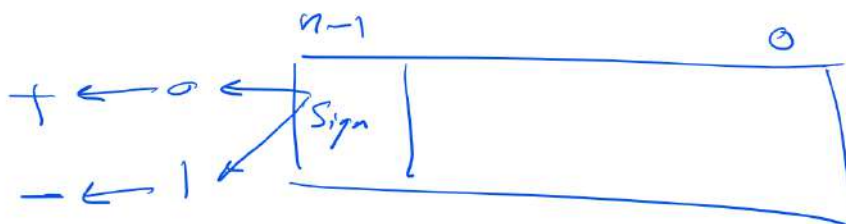
if $n = 3$ what is the range of values in decimal



if $n = 4$ what is the range of values in binary



Signed Numbers



(6)

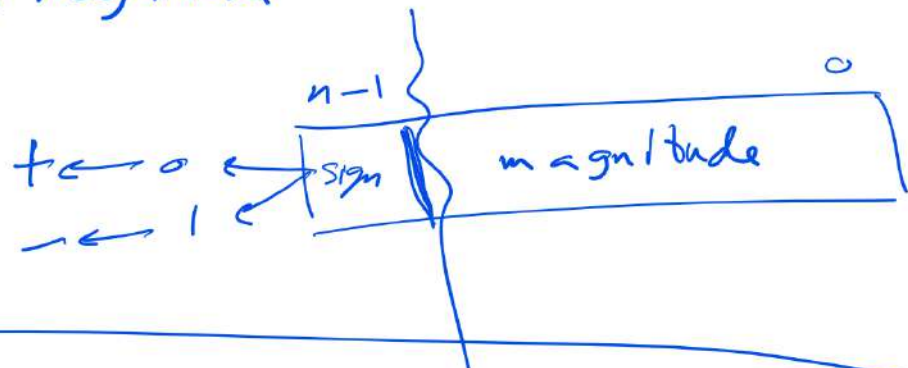
- 1- sign and magnitude
- 2- 1's complement (first)
- 3- 2's complement (second)

n bits

Addition in binary

$$\begin{array}{r}
 \cancel{000} \\
 \hline
 \begin{array}{r}
 1 \\
 + 1 \\
 \hline
 0 \\
 1 \\
 1
 \end{array}
 \end{array}
 \quad
 \begin{array}{r}
 1 \\
 + 1 \\
 \hline
 0 \\
 1 \\
 0
 \end{array}
 \quad
 \begin{array}{r}
 0 \\
 + 1 \\
 \hline
 0 \\
 1 \\
 1
 \end{array}
 \quad
 \begin{array}{r}
 0 \\
 + 1 \\
 \hline
 0 \\
 0 \\
 0
 \end{array}$$

(1) sign and magnitude

if $n = 4$ bits

$$+6 = \boxed{0 \mid 1 \mid 1 \mid 0} \quad \leftarrow$$

Diagram showing the binary representation of +6 in a 4-bit sign and magnitude format. The sign bit is 0, and the magnitude is 110. An arrow points to the magnitude part.

$$+6 = (0110)_2$$

$$-6 = 1110 \quad \leftarrow$$

Diagram showing the binary representation of -6 in a 4-bit sign and magnitude format. The sign bit is 1, and the magnitude is 110. An arrow points to the magnitude part.

(7)

if $n = \underline{8 \text{ bit}}$

$$-6 = 1000 \overset{1}{0} \overset{2}{1} \overset{1}{1} 0$$

* Range of values in sign and magnitude:

$$-(2^{n-1}-1) \longrightarrow +(2^{n-1}-1)$$

Example: if $n = 4 \text{ bits}$

$$\underline{-(2^{4-1}-1)} \longrightarrow +(2^{4-1}-1)$$

$$-7 \longrightarrow +7$$

$$1111 \longrightarrow 0111$$

[2] 1's complement:

if $n = \underline{4 \text{ bits}}$

$$+6 = ?$$

$$+6 = 0110$$

$$-6 = ? \longrightarrow +6 = 0110$$

$$-6 = (1001)_2$$

$$-6 = 1001$$

1's

(8)

if $n = 8 \text{ bits}$

$-6 = ?$

$$+6 = (00000110)_2$$

$\downarrow$
 1's

$$\underline{11111001}$$

$-6 = (11111001)_2$

Range of values in 1's complement

$$-(2^{n-1} - 1) \longrightarrow + (2^{n-1} - 1)$$

Example: if $n = 4 \text{ bits}$ Range of values:

$$-(2^{4-1} - 1) \longrightarrow + (2^{4-1} - 1)$$

$$-7 \longrightarrow +7$$

$$\underline{1000} \longrightarrow \underline{0111}$$

[3] 2's complement:if $n = 4 \text{ bits}$

$+6 = 0110$

$-6 = ?$

$$+6 = 0110$$

$\downarrow$
 1's

$$\underline{1001}$$

$$+ \quad \underline{1001}$$

$$\hline 1010$$

$-6 = (1010)_2$

(9)

if $n=8\text{ bits}$ $-6=?$

$$\begin{array}{r}
 +6 = 0000\ 0110 \\
 + \quad 1111\ 1001 \\
 \hline
 1111\ 1010
 \end{array}$$

Diagram illustrating the addition of 6 and its 2's complement to find -6. The 2's complement of 6 (0110) is 1001. The sum of 0110 and 1001 is 1010. The final result is 1111 1010.

$$-6 = (1111\ 1010)_2$$

لوجد طريقة سرية N 2's

$$\begin{array}{r}
 0000\ 0110 \\
 \downarrow \\
 1111\ 1010
 \end{array}$$

Diagram illustrating the 2's complement operation. The 2's complement of 0110 is 1010. The final result is 1111 1010.

Range of values in 2's complement

$$-(2^{n-1}) \longrightarrow +(2^{n-1}-1)$$

Example : if $n=4\text{ bits}$

$$-(2^{4-1}) \longrightarrow +(2^{4-1}-1)$$

$$-8 \longrightarrow +7$$

$$1000 \longrightarrow 0111$$

what is the decimal value for $(\underline{001010})_2$ and $(\underline{110011})_2$. If these numbers are represented as (11)

- ① sign and magnitude
- ② 1's complement
- ③ 2's complement.

① $(001010)_2 = +10$

$(11^4 00^{21})_2 = -19$

② $(00^{8421} 1010) = +10$

$(\underline{110011}) = -12$

$\begin{matrix} 8 & 4 & 2 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{matrix} \xrightarrow{1's} 201100 = +12$

③ $(00^{8421} 1010)_2 = +10$

$(110011) \xrightarrow{2's} 001101$

$(110011) = -13$

what is the value of -17 in the following (12)

1- sign and magnitude

2- 1's complement

3- 2's complement

use 8 bits

① sign and magnitude

$$\begin{array}{r} 168421 \\ -17 = 10010001 \end{array}$$

② 1's complement

$$-17 = 3$$

$$-17 = 11101110$$

$$\begin{array}{r} 168421 \\ +17 = 00010001 \\ -17 = 11101110 \end{array}$$

③ 2's complement

$$-17 = 11101111$$

$$\begin{array}{r} +17 = 00010001 \\ 11101111 \end{array}$$

$$* \text{ 2's complement} = \text{1's complement} + 1$$

$K_1 = (2^n - 1) - P$ 1's complement

$K_2 = 2^n - P$ 2's complement

$n=4$
 $+6 = 0110$
 $-6 = 1001$
1's
 $K_1 = (2^4 - 1) - 6$
 $= 9 = 1001$

$P = \text{positive value}$

Addition

① Sign and magnitude

- when the two numbers have the same sign use one of them
then add numbers

$n = 6 \text{ bit}$

$+3$
 $+4$

 $+7$

000011
 $+ 000100$

 000111

- when the two numbers have different sign. use the sign of larger number. then subtract.

$n = 6$ bits

(13)

$$\begin{array}{r} -5 \\ +3 \\ \hline -2 \end{array}$$

$$\begin{array}{r} \textcircled{1}00101 \\ + 00011 \\ \hline 100010 \end{array}$$

② 1's Complement

if there is carry at the end you have to add it.
to the result.

$n = 8$ bits

$$\begin{array}{r} 11101100 \\ + 11000001 \\ \hline 10101101 \\ \text{+} \rightarrow 1 \\ \hline 10101110 \end{array}$$

③ 2's Complement

if there is carry at the end you have to ignore it.

$n = 8$ bits

$$\begin{array}{r} 11100010 \\ + 11000001 \\ \hline 10100011 \end{array}$$

← $\text{Carry} = 1$

In 2's Complement,

(14)

overflow: when the result does not fit in the range $\rightarrow$ arithmetic overflow

OV = 1

- ① if there is carry from D_{n-2} to D_{n-1} and no carry from D_{n-1} to out.
- ② if there is carry from D_{n-1} to out and no carry from D_{n-2} to D_{n-1}

$$\begin{array}{r} C_7 \\ \textcircled{1} \\ 11000100 \\ + 11001001 \\ \hline 10001101 \\ \swarrow \\ C_8 = \textcircled{1} \\ OV = 0 \end{array}$$

$$\begin{array}{r} \textcircled{0}111 \\ 10111010 \\ + 10101000 \\ \hline 01100010 \\ \swarrow \\ C_8 = \textcircled{1} \\ OV = 1 \\ OV = C_8 \oplus C_7 \end{array}$$

$$OV = C_n \oplus C_{n-1}$$

$$\begin{aligned} &= 1 \oplus 0 \\ &= 1 \end{aligned}$$

Q3.5

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